

■ SQL Joins & PySpark Joins — Study Notes

■ 1■■ INNER JOIN

Returns only matching records from both tables.

SQL Example:

```
SELECT t1.*, t2.*
FROM table1 t1
INNER JOIN table2 t2
    ON t1.column = t2.column;
```

PySpark Example:

```
result = df1.join(df2, df1.column == df2.column, 'inner')
```

■ 2■■ LEFT (OUTER) JOIN

Returns all records from the left table, and matching records from the right. Non-matching right-side rows have NULLs.

SQL Example:

```
SELECT t1.*, t2.*
FROM table1 t1
LEFT JOIN table2 t2
    ON t1.column = t2.column;
```

PySpark Example:

```
result = df1.join(df2, df1.column == df2.column, 'left')
```

■ 3■■ RIGHT (OUTER) JOIN

Returns all records from the right table and matching records from the left.

SQL Example:

```
SELECT t1.*, t2.*
FROM table1 t1
RIGHT JOIN table2 t2
    ON t1.column = t2.column;
```

PySpark Example:

```
result = df1.join(df2, df1.column == df2.column, 'right')
```

■ 4■■ FULL (OUTER) JOIN

Returns all records from both tables, matching where possible. Missing values are filled with NULL.

SQL Example:

```
SELECT t1.*, t2.*
FROM table1 t1
FULL JOIN table2 t2
    ON t1.column = t2.column;
```

PySpark Example:

```
result = df1.join(df2, df1.column == df2.column, 'full')
```

■ 5 ■ CROSS JOIN

Returns Cartesian product of both tables — every row from t1 is paired with every row from t2.

SQL Example:

```
SELECT t1.*, t2.*
FROM table1 t1
CROSS JOIN table2 t2;
```

PySpark Example:

```
result = df1.crossJoin(df2)
```

■ Equi Join vs Non-Equi Join

Type	Definition	Example
Equi Join	Join condition uses equality (=) between columns.	t1.id = t2.id
Non-Equi Join	Join condition uses non-equality operators like <, >, BETWEEN	t1.salary BETWEEN t2.min_salary A

■ PySpark Joins Summary

Join Type	Description	Equivalent SQL Logic
inner	Returns matching records from both DataFrames.	INNER JOIN
left / left_outer	Returns all records from left DF and matched ones from right.	LEFT JOIN
right / right_outer	Returns all records from right DF and matched ones from left.	RIGHT JOIN
full / full_outer	Returns all records from both sides.	FULL JOIN
cross	Cartesian product of two DFs.	CROSS JOIN
left_semi	Returns only left-side rows with at least one match in right DF (no right-side rows).	Similar to EXISTS
left_anti	Returns only left-side rows with no match in right DF.	Similar to NOT EXISTS

■ Example: Left Semi & Left Anti Joins

****Left Table (A):****

id
1
1

****Right Table (B):****

id
1
1
1

Join Type	Output Rows	Explanation
Inner Join	6 rows	Each left 1 matches with all three right 1s → $2 \times 3 = 6$
Left Semi Join	2 rows	Only left-side rows having at least one match (no duplication)
Left Anti Join	0 rows	No unmatched rows in A (all found in B)